

DepthWizard

Complete Technical Handover • SIH 26175 • Single-View Height Estimation and 3D Flythrough
ISRO / Space Applications Centre • Disaster Management

How to use this document

This is written so a new engineer, or a new AI coding assistant with no prior context, can continue this project without repeating work. It explains not only what was built but why each decision was taken, and critically, which approaches were tried and failed.

Read Section 8 (Dead Ends) before writing any code. Roughly two days of development were spent discovering that certain approaches cannot work in this environment. Repeating them will cost the same time again.

CLAUDE.md, STATUS.md and docs/HANDOVER.md in the repository are authoritative for current implementation detail. This document supplies the reasoning and history those files do not capture.

1. What the project is

The problem statement asks for software converting a single 2D optical image (satellite or aerial; PNG, JPG or GeoTIFF) into an accurate Digital Surface Model, letting a user navigate the reconstructed terrain in real time through an interactive 3D flythrough. It sits under Disaster Management because elevation drives flood screening, landslide risk assessment and emergency route planning — and conventional sources take days to weeks.

Evaluation is 50/50. Half is DSM accuracy: RMSE, MAE and correlation against LiDAR or reference data, with stability required across urban, sparse, hilly and forested landscapes. Half is rendering and user experience: projection accuracy, visual fidelity, flythrough navigability, interface intuitiveness, software stability and standalone deployment.

Two required input paths. A PNG or JPG without georeferencing must produce a **relative** DSM with units explicitly null. A GeoTIFF must produce an **absolute metric** DSM, calibrated using a low-resolution DEM such as SRTM 30 m or minimal ground control points. The official SAC reference is github.com/IMG-PROCESS-SAC/SIH2026/ and recommends GAMUS.

2. The central technical insight

This is the most important thing to understand about the project.

The obvious approach, taken by essentially every other submission, is to feed the image to a monocular depth foundation model such as Depth Anything V2 or MiDaS, then scale the relative depth against a reference DEM. **This does not work on nadir satellite imagery, and we have measured evidence of why.**

Monocular depth models learn from ground-level photography. Their cues are perspective convergence, occlusion ordering, vanishing-point geometry and familiar object scale. A top-down satellite tile contains almost none of these. Deprived of them the model falls back on **local texture variance**: tree canopies, being visually noisy, become raised blobs, while flat uniform rooftops collapse into the base plane. The result looks plausible and means nothing.

Measured: Depth Anything V2 calibrated against reference elevation at $R^2 = 0.099$. MiDaS Small achieved a brightness-to-height correlation of 0.17; DAv2 reached 0.64 — but that correlates with *brightness*, which is what the model actually keys on, not with true height.

Published literature confirms this independently. arXiv:2604.02009 states that near-orthographic viewing geometry eliminates perspective height cues, forcing reliance on shadow information, and that illumination variation can cause height errors exceeding 30 percent. The problem statement's own background says the same in different words.

The conclusion the project is built on: metric height from a single nadir image comes from photogrammetry, not a learned depth prior. Specifically from shadow geometry, $h = L \times \tan(\theta_{\text{sun}})$, where L is shadow length in metres and θ is solar elevation. This is measurement, not inference.

3. Architecture

3.1 Pipeline

```
single-acquisition image (+ solar metadata if present)
|
+-- GeoTIFF detection (rasterio)
|   CRS, affine transform, GSD, nodata
|
|   no georeferencing -> relative path, units = null
```

```

|
+-- SAM instance segmentation (default 512px / 12 points)
|   watershed splitting of oversized masks
|   shape-based classification: building | tree | road | ground
|
+-- shadow detection and matching
|   shadow blobs paired to building instances
|   h = L * tan(theta_sun)
|
+-- multi-cue height fusion
|   inverse-variance weighted across 5 methods
|   disagreement -> low-confidence flag, never silent averaging
|
+-- quality gates
|   coherence R, azimuth agreement, Rayleigh p
|
+-- scene assembly
|   buildings -> flat-top extruded prisms (LOD1) at metric height
|   roads / ground -> flat polygons, source imagery draped
|   trees -> instanced low-poly canopy (toggleable)
|   export -> GeoTIFF float32 + sidecar JSON

```

3.2 Why LOD1 extrusion, not a heightfield

A heightfield stores one height per grid cell and the renderer interpolates between neighbouring vertices. It is **mathematically incapable of producing a vertical wall**. Early versions used a displaced plane and produced stretched streaks where a rooftop vertex sat beside a ground vertex, and blobby merged mounds instead of distinct buildings.

Buildings are flat-topped prisms. LOD1 imposes that prior: each footprint becomes its own mesh with a flat triangulated roof at height h and vertical wall quads extruded to ground, sharing no vertices with the ground plane or with neighbours. Sharp edges are then correct by construction.

3.3 The five height cues

Method	Principle	Reliability	Fails when
A. Shadow geometry	$h = L \times \tan(\theta)$	Primary — physics	Shadow occluded; overcast; near-overhead sun
B. Facade visibility	Visible wall extent, off-nadir	High where present	True-ortho products
C. Relief displacement	Roof offset from footprint $\propto$ height	Moderate	Orthorectified imagery
D. Depth-calibrated	DAv2 ordering scaled to shadow anchors	Low — ordering only	Calibration R^2 below threshold
E. Neighbourhood prior	Median of confident neighbours	Lowest	Heterogeneous blocks

Fusion is inverse-variance weighted, with uncertainty floored at 0.5 m to avoid division by near-zero. Method D is conceptually important: it rescues the depth model by using it only for what it is good at — relative ordering — calibrated against buildings where shadow measurement succeeded. It is admitted only if that calibration passes an R^2 threshold, which in testing it did not.

3.4 Solar geometry priority chain

Priority	Source	Notes
1	Product metadata: .IMD, .XML, .MTL, EXIF sidecar	Exact. Parsed by sun_geometry.py
2	Computed from datetime + scene lat/lon via pysolar	Exact orbital geometry
3	Azimuth measured from the image's own shadow offsets	Cross-check; gated, see Section 4
4	User-supplied slider	Labelled 'estimated' in the UI

Critical correctness rule: when measured shadows win, they supply only the azimuth. Elevation must be inherited from metadata, computed or user values — never a default. An earlier bug used a hardcoded 45-degree default here, understating every height by $\tan(74)/\tan(45) = 3.49$ on the Kathmandu dataset. This class of bug is silent and survives into presentations unnoticed.

4. The quality gates

These are the project's main differentiator. The system refuses to claim metric heights it cannot justify, and states why.

4.1 Relative versus metric

No georeferencing produces a dimensionless rDSM with units explicitly null. Relative values are never labelled as metres, anywhere. A persistent provenance badge shows metric or relative status, calibration source and estimated uncertainty at all times.

4.2 Shadow coherence

Sun azimuth is measured independently from every building's shadow offset direction. If those directions disagree, the detected shadows are not real cast shadows. Coherence R is the resultant length of those direction vectors.

Pairs are weighted by **offset squared**, because a longer shadow carries more angular information than a short one. Introducing this weighting improved crop1's azimuth error from 56.7 degrees to 5.4 degrees — strong evidence the estimator is correct rather than tuned to pass a threshold.

4.3 Two-tier admission

Coherence alone is insufficient: a high R can arise from consistently detecting the same *artefact* — roof edges, awnings or tree shade aligned with a street grid. Admission therefore requires both a statistical and a physical test.

Tier	Condition	Meaning
MEASURED	$R \geq 0.5$ AND azimuth delta $\leq 10^\circ$	Certified metric
BORDERLINE	Rayleigh $p < 0.01$ AND tight azimuth agreement	Physically consistent, coherence marginal
REJECTED	Either test fails	Downgraded to relative, reason logged

The proof both gates are needed is Antakya crop4: statistically significant by Rayleigh ($p = 0.004$) with 29 pairs pointing consistently, yet an azimuth delta of 48.7 degrees blocks it. A statistical test alone would have admitted a systematic artefact. Significance proves the pattern is not random; physical agreement proves it is the sun.

4.4 Mosaic detection

Composite basemaps stitch acquisitions from different dates and times, so solar geometry is undefined across the image. Detected via low coherence and reported distinctly from 'no shadows found'. See 8.1 for why this matters enormously.

4.5 Feasibility pre-check

$$h_{\min} = \min_offset_px * GSD * \tan(\theta_{\text{sun}})$$

The minimum building height that can produce a measurable shadow offset. Reported before processing; a warning fires above 60 degrees sun elevation. Derived from first principles and then validated against two datasets:

Site	Sun elevation	GSD	h min	Outcome
Antakya	28.3°	0.305 m	0.82 m	Method validated; heights credible
Kathmandu	74.0°	0.500 m	28 m	Most buildings below floor; unusable

4.6 Coverage reporting

Three categories, always separate, never one percentage: **directly measured** (shadow, facade, lean), **inferred** (depth-calibrated or neighbour prior), and **failed**. An earlier version reported 100 percent coverage, which was a fallback silently filling gaps. That was removed.

5. Code structure

Reconstructed from development history; verify against the repository, which is authoritative.

File	Responsibility
app/main.py	FastAPI app. POST /upload, POST /estimate-heights, GET /health, GET /progress/{id} (SSE), GET /demo/list, GET /demo/{file}. Non-blocking model load in a background thread.
app/estimators/base.py	HeightEstimator abstract base class and HeightMetadata. The swap interface: estimate(image_bytes) -> (array, metadata).
app/estimators/synthetic.py	Gaussian hills plus box buildings. Scaffold for testing the render path with no model. Keep it — it is the fallback when models fail.
app/estimators/midas.py	MiDaS Small via torch.hub. Requires a trust bypass, see 8.4.
app/estimators/depth_anything.py	Depth Anything V2 Small from a LOCAL directory via HF_DEPTH_MODEL_PATH. Must never hit the network on the dev machine.

File	Responsibility
app/sam_segmentation.py	SAM mask generation, watershed splitting of oversized masks, building area histogram logging.
app/height_estimation.py	Sun priority chain, per-building height assembly, three-category coverage, per-building diagnostic table.
app/height_fusion.py	The five estimation methods and inverse-variance fusion with pairwise disagreement checks.
app/sun_geometry.py	Metadata sidecar parsing, pysolar computation, shadow-azimuth measurement, <code>cross_check_sun_sources()</code> , <code>SunSourceLog</code> .
app/depth_postprocess.py	High-pass detrending removing the depth model's bogus low-frequency ramp; Gaussian edge taper.
app/structure_dsm.py	SLIC superpixels, shape-based classification, outlier clamping, union-find merge of adjacent building segments.
app/calibration.py	GeoTIFF detection, RANSAC / Huber robust fitting, SRTM-based scale calibration.
app/validation.py	MAE / RMSE / R^2 / bias against a reference DSM.
app/tiling.py	Overlapping tile split with cosine-feathered blend for arbitrary input size.
eval/gamus.py	GAMUS evaluator. Two protocols (frozen global, oracle). Five sanity tests passing. BLOCKED: awaits paired .h5 tiles in <code>data/gamus/</code> .
eval/osm_reference.py	OSM height validation via Overpass API. Functional; produced zero matches, see 8.6.
download_models.py	Downloads model weights at container startup. Used by the Docker deployment.
Dockerfile	HF Spaces deployment. Non-root user, port 7860, health check with generous start-period.
static/index.html	Three.js frontend: LOD1 renderer, four camera modes, measurement, cross-section, flood slider, provenance badge, colour ramp legend, SSE progress, tree toggle, <code>?demo=</code> autoload.
static/landing.html	Landing page with demo scene buttons.

6. Models used

Model	Role	Size	Source
Segment Anything (ViT-B)	Instance segmentation producing building and tree masks	~375 MB	Meta
Depth Anything V2 Small	Relative depth for ordering only; rejected when calibration fails	99 MB	Local dir via <code>HF_DEPTH_MODEL_PATH</code>
MiDaS Small	Earlier depth backend, retained for comparison	82 MB	torch.hub (GitHub releases)

No model was fine-tuned. All are zero-shot. Fine-tuning Depth Anything V2 on GAMUS nDSM labels is the highest-value roadmap item: the literature reports up to 24 percent RMSE reduction from exactly this adaptation, and Depth2Elevation reports up to 42 percent relative improvement on GAMUS.

No neural rendering. NeRF and Gaussian splatting were evaluated and deliberately rejected: the problem statement names raster engines, and Three.js rasterisation meets the requirement at a fraction of the compute cost.

7. Results achieved

7.1 Validation dataset

Antakya, Turkey. WorldView pre-earthquake imagery from the Vantor / Maxar Open Data Program (CC-BY 4.0). GSD 0.305 m, sun elevation 28.3 degrees. Four 1024 x 1024 crops, selected for building density. Pre-event imagery was used deliberately so the building stock matches historical records — Antakya was devastated by the February 2023 earthquake.

7.2 Per-crop results

Crop	R (weighted)	Azimuth Δ	Rayleigh p	Tier	Buildings
crop1	0.490	5.4°	0.0017	BORDERLINE	12 borderline, 0 failed
crop2	0.681	10.6°	—	MEASURED	22 of 23, 0 failed
crop3	0.275	40.6°	0.062	REJECTED	19 relative, 2 failed
crop4	0.384	48.7°	0.004	REJECTED	29 relative, 1 failed

- Totals: 88 buildings, 34 measured or borderline, 48 relative, 2 failed

- crop2: mean 4.6 m, median 2.8 m, range 1.8 to 14.7 m
- crop1: mean 4.3 m, median 2.4 m — independently consistent with crop2
- Measured sun azimuth 172.6° versus computed 162.0° — agreement within 10.6° between two entirely independent methods
- Heights consistent with Antakya's known 1 to 2 storey Ottoman stone housing

7.3 Performance

Configuration	Detection quality	Runtime
SAM 384 / 8 points	Lower recall	23.8 s
SAM 512 / 12 points (default)	96% centroid recall, 9.4% mask IoU	65 s

Reduced from an initial 536 s by: vectorising shadow matching with a single pre-dilated label map instead of per-building dilations; tuning the SAM prompt grid; batching tile inference; vectorising shape features with `scipy.ndimage`; capping SLIC iterations at 5 (3.4 s to 1.0 s, no visible quality loss); and moving blocking inference to a worker thread so SSE progress streams concurrently. CPU only, approximately 4 GB RAM.

Demo scenes are precomputed and render in under two seconds. The UI labels these as precomputed so the fast render is never mistaken for pipeline speed.

7.4 What is NOT achieved — be honest about this

- **No RMSE, MAE or R^2 against ground truth.** No reference DSM existed for any validation site. All results are internal consistency and physical plausibility, not measured error. This is the largest remaining gap and it is half the rubric. `eval/gamus.py` is built with five passing sanity tests and needs only paired tiles.
- **Footprint boundary precision.** 96 percent centroid recall but 9.4 percent mask IoU. Buildings are located but boundaries are imprecise in dense fabric. Always quote both figures together — the IoU alone is misleading.
- **Systematic height underestimate.** Median 3.4 m against an expected 3 to 3.5 m per storey. Root cause identified: shadow truncation. The mask shadow_candidate `&= ~all_buildings` removes shadows falling on neighbouring rooftops, shortening measured shadow length.
- **Crops were selected for building density**, so coverage is not scene-wide.
- **Landscape breadth untested.** The rubric requires stability across urban, sparse, hilly and forested; only urban has been evaluated.

8. Dead ends — do not repeat these

The most valuable section of this handover. Each item consumed significant time and none can be made to work in this environment.

8.1 Google Maps screenshots are unusable for shadow photogrammetry

The single most important finding. Public basemap imagery is a **mosaic**: Google documents that aerial images are composited from several images taken over multiple days or months, and the acquisition date changes as the cursor moves across the map.

Different parts of one screenshot therefore have **different sun positions**. Shadow coherence measured 0.02 to 0.36 on such input — near-random, and correctly so. Shadow photogrammetry is physically impossible on a mosaic because there is no single sun. Roughly a day was lost debugging a pipeline that was working correctly on unusable data.

Always use single-acquisition imagery. Vantor / Maxar Open Data provides free single-acquisition Cloud Optimized GeoTIFFs at 30 to 50 cm, CC-BY 4.0. Browse at opengeos.org/maplibre-gl-vantor. A path like `.../104001009C0CE600-visual.tif` is one catalog ID, one capture, one sun position.

8.2 Select imagery by sun elevation, not cloud cover

The first Vantor download was chosen for lowest cloud cover (15 percent) and proved to be a 74-degree sun — nearly overhead, giving $h_{\min} = 28$ m. Most Kathmandu buildings fell below the detection floor and the pipeline produced implausible 2 to 112 m heights. **Target 25 to 50 degrees:** winter acquisitions, or early morning and late afternoon captures. Compute sun elevation from the candidate's date, time and coordinates before downloading.

8.3 Hugging Face is blocked on the development machine

- huggingface.co is blocked in the browser (corporate network).

- From Python the HF API responds 200, but the **Xet CDN** (us.aws.cdn.hf.co / cas-server.xethub.hf.co) returns persistent **503** on weight downloads.
- HF_HUB_DISABLE_XET=1 is **not honoured** — tested on huggingface_hub 0.29.3 and current. The repo serves Xet URLs regardless of client version.
- HF_ENDPOINT=https://hf-mirror.com resolves but returns the wrong file structure (no safetensors for this model).
- **Working solution:** download weights on an unrestricted device, push to a private GitHub repository, clone on the work machine, and point the loader at the local directory via HF_DEPTH_MODEL_PATH. GitHub is reachable.

Note: a Hugging Face Space can reach huggingface.co even though the laptop cannot, so for deployment the weights download at container startup.

8.4 torch.hub trust prompt blocks MiDaS

MiDaS_small internally calls torch.hub.load for rwrightman/gen-efficientnet-pytorch, triggering an interactive trust prompt that cannot be answered from a script. trust_repo=True on the outer call does not help, and writing the trusted_list file does not either. The working bypass:

```
import torch.hub
torch.hub._check_repo_is_trusted = lambda *a, **k: None
```

MiDaS also requires timm and opencv (use opencv-python-headless).

8.5 WebGL float textures are unavailable on this machine

OES_texture_float is not supported, so float32 height data was silently clamped to 0–1 and every terrain rendered flat. The fix, which is also more portable and should be kept: pack heights as **16-bit across the R and G channels** of an RGBA Uint8 texture.

```
texture = new THREE.DataTexture(rgba, w, h,
    THREE.RGBAFormat, THREE.UnsignedByteType);
texture.minFilter = THREE.NearestFilter; // MANDATORY
texture.magFilter = THREE.NearestFilter; // bilinear corrupts packed bytes
// unpack in shader:
// float v = (t.r * 255.0 * 256.0 + t.g * 255.0) / 65535.0;
```

NearestFilter is not optional. Bilinear filtering interpolates R and G as independent colour channels and corrupts the packed value at texel boundaries.

8.6 Validation routes that failed

- **OpenStreetMap heights, Antakya:** of 324 mapped buildings across four crops, **zero** had a height tag and only 8 had building:levels. Turkish cities are well mapped for footprints but not height. Zero matches resulted. eval/osm_reference.py is functional and would work at a densely annotated site — Munich, Amsterdam or Paris.
- **ISPRS Potsdam:** requires a data request form with *manual approval*, too slow for a deadline. Worth submitting early: it provides orthophoto plus DSM plus normalised DSM at 5 cm GSD, exactly what an RMSE evaluation needs.

9. Development environment

Item	Value
Machine	Corporate Windows laptop with network restrictions
Project root	C:\Users\rtadi001\ClaudeAI\depthwizard
Python	3.12, per-user install under %LOCALAPPDATA%\Programs\Python
Repository	github.com/AKHI1630/depthwizard
Weights transfer repo	github.com/AKHI1630/my-project (Depth Anything V2 files)
Dev server port	8002 (8000 and 8001 held by stale sockets)
Container port	7860 (Hugging Face Spaces requirement)

Practical notes. Do not run uvicorn with --reload: it watches .venv and restarts the server during pip installs. Clear stale processes with Get-Process python | Stop-Process -Force before starting. Keep the project out of OneDrive-synced folders — sync locks files mid-write and breaks installs.

10. Working principles — keep these

- **No silent catches.** Every failure logs with context and surfaces a visible UI message. A swallowed `RangeError` once cost an hour of debugging a flat plane.
- **Never substitute a default that yields a plausible-looking wrong result.** Fail loudly. The 45-degree elevation default that understated every height by 3.5x is the cautionary example.
- **The stats bar always names which pipeline ran.** Three separate times the old heightfield path rendered while the new metric path had silently failed, and it was not obvious from the screen.
- **Sanity-test any evaluation harness before trusting a metric.** Feeding the reference as prediction must give exactly zero error; a constant prediction must give a hand-checkable baseline.
- **Verify sign conventions empirically, never by reasoning.** Log mean final height over bright rooftop pixels versus dark road pixels and assert rooftops are higher.
- **No timed auto-redirects in the UI.** They race against the user's own click.

11. Roadmap — ordered by value

11.1 Highest priority: get an RMSE number

Half the rubric, currently empty. `eval/gamus.py` is built and its sanity tests pass; it needs data. Obtain a handful of GAMUS tiles (huggingface.co/datasets/earthflow/GAMUS) via the GitHub transfer method in 8.3. Files are HDF5, one per tile, named like `DC_04_23_RGB.h5`, with paired height maps derived from LiDAR. The official loader at github.com/EarthNets/RSI-MMSegmentation documents the internal key names — read it rather than guessing.

Report two protocols: a **frozen global affine** fitted on validation and applied unchanged at test time with zero ground-truth access (the honest deployment number), and an **oracle** fitted per crop (theoretical ceiling, never presented as deployment accuracy). Note the domain shift honestly: GAMUS is US aerial orthophotos, validation so far was Turkish satellite imagery.

Fallback if the transfer is impossible: synthetic validation. Generate scenes with boxes of known heights and geometrically exact shadows, then measure recovery error. Not real-world accuracy, but a genuine measurement of the algorithm, requiring no download.

11.2 Footprint boundary quality

9.4 percent mask IoU is the most visible weakness. Footprints are blobby because SAM and SLIC follow colour, not building boundaries. Real buildings are rectilinear. Regularisation: `minAreaRect` per component, and where component area divided by rectangle area exceeds about 0.6, replace the blob with the rectangle. Otherwise simplify with `approxPolyDP` and snap edge orientations to the dominant scene angle, since buildings in a block usually share orientation.

11.3 Correct the shadow truncation bias

Median 3.4 m is systematically low because shadows falling on neighbouring rooftops are masked out. Options: allow shadow candidates on rooftops when the rooftop is lower than the casting building; or model the truncation and correct for it; or restrict measurement to buildings with unobstructed shadow paths and report reduced coverage honestly.

11.4 Further

- Fine-tune Depth Anything V2 on GAMUS nDSM as a shadow-independent fallback
- Evaluate across all four required landscape types, not just urban
- Relief displacement (method C) for off-nadir products
- Standalone desktop packaging (Electron or Tauri) for the deployment requirement

12. First actions for whoever picks this up

- Read `CLAUDE.md`, `STATUS.md` and `docs/HANDOVER.md` in the repository.
- Read Section 8 of this document before writing any code.
- Start the server, load the Antakya crop2 demo scene, and confirm you reproduce 22 of 23 buildings measured at $R = 0.681$. If not, something has regressed.
- Then work Section 11 in order. **Do not re-architect.** The current design is the product of three rewrites driven by measurement; the returns on a fourth are negative.